

MATH 114 HW 3

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Question 1

a.

$$\begin{aligned}\mathbb{P}(X_0 = 0, X_1 = 1, X_2 = 0) &= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{12} \\ \mathbb{P}(X_0 = 0, X_2 = 0) &= \mathbb{P}(X_0 = 0, X_1 = 0, X_2 = 0) \\ &\quad + \mathbb{P}(X_0 = 0, X_1 = 1, X_2 = 0) \\ &\quad + \mathbb{P}(X_0 = 0, X_1 = 2, X_2 = 0) \\ &= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{12} + 0 \\ &= \frac{5}{24}\end{aligned}$$

b.

$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \\ 0 & \frac{1}{4} & \frac{3}{4} \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$\frac{1}{2}a + \frac{1}{3}b = a$$

$$\frac{1}{2}a + \frac{1}{2}b + \frac{1}{4}c = b$$

$$\frac{1}{6}b + \frac{3}{4}c = c$$

$$\frac{1}{3}b = \frac{1}{2}a$$

$$\frac{2}{3}b = a$$

$$\frac{1}{6}b = \frac{1}{4}c$$

$$\frac{2}{3}b = c$$

$$a + b + c = 1$$

$$\frac{2}{3}b + b + \frac{2}{3}b = 1$$

$$\frac{7}{3}b = 1$$

$$b = \frac{3}{7}$$

$$a = c = \frac{2}{7}$$

$$\pi = \begin{bmatrix} \frac{2}{7} & \frac{3}{7} & \frac{2}{7} \end{bmatrix}$$

c.

$$\begin{aligned}
\lim_{n \rightarrow \infty} \pi_0 P^n &= \pi_0 \lim_{n \rightarrow \infty} P^n \\
&= \pi_0 \begin{bmatrix} \pi \\ \vdots \\ \pi \end{bmatrix} \\
&= \left(\pi(1) \cdot \sum_i \pi_0(i), \pi(2) \cdot \sum_i \pi_0(i), \dots \right) \\
&= \pi
\end{aligned}$$

Question 2

Let $S = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$ be $\{1, 2, 3, 4\}$

Then we have

$$p_{1,1} = \mathbb{P}(Y_2 = (0, 0) | Y_1 = (0, 0)) = \mathbb{P}(X_2 = 0 | X_1 = 0) = \alpha$$

Continuing this, we have

$$\begin{bmatrix} \alpha & 1 - \alpha & 0 & 0 \\ 0 & 0 & 1 - \beta & \beta \\ \alpha & 1 - \alpha & 0 & 0 \\ 0 & 0 & 1 - \beta & \beta \end{bmatrix}$$

Question 3

Since only coordinate can change at a time uniformly at random,

$$p(\mathbf{y}, \mathbf{z}) = \begin{cases} \frac{1}{m} & \text{if } \mathbf{y} \text{ and } \mathbf{z} \text{ differ in exactly one coordinate} \\ 0 & \text{otherwise} \end{cases}$$

Question 4

a. $p(i, j)$ = probability of moving from i to j , since it's a probability, it must be non-negative.

$\sum_{j=1}^N p(i, j)$ = sum of probabilities of each destination = 1, therefore each row of P sums to 1.

$$\text{b. } P \cdot \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} p_{11} + \dots + p_{1N} \\ \vdots \\ p_{N1} + \dots + p_{NN} \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

c.

$$\begin{aligned}\mathbb{P}((A \cap B) \cap C) &= \mathbb{P}(A \cap B|C) \cdot \mathbb{P}(C) \\ \mathbb{P}(A \cap (B \cap C)) &= \mathbb{P}(A|B \cap C) \cdot \mathbb{P}(B \cap C) \\ &= \mathbb{P}(A|B \cap C) \cdot \mathbb{P}(B|C) \cdot \mathbb{P}(C)\end{aligned}$$

Setting these equal yields

$$\mathbb{P}(A \cap B|C) = \mathbb{P}(A|B \cap C) \cdot \mathbb{P}(B|C)$$

$$\begin{aligned}p_{n+1}(i, j) &= \mathbb{P}(X_{n+1} = j|X_0 = i) = \sum_{k \in S} \mathbb{P}(X_{n+1} = j, X_n = k|X_0 = i) \\ &= \sum_{k \in S} \mathbb{P}(X_{n+1} = j|X_n = k, X_0 = i) \cdot \mathbb{P}(X_n = k|X_0 = i) \\ &= \sum_{k \in S} \mathbb{P}(X_{n+1} = j|X_n = k) \cdot \mathbb{P}(X_n = k|X_0 = i) \\ &= \sum_{k \in S} p(k, j) \cdot p_n(i, k) \\ &= \sum_{k \in S} p_n(i, k) \cdot p(k, j) \\ &= [P_n P]_{i, j}\end{aligned}$$

d.

$$\begin{aligned}\pi_n &= \pi_0 P^n \\ \pi_0 P^n &= \pi_0 P \cdot P^{n-1} \\ &= \pi_1 P^{n-1} \\ &= \pi_2 P^{n-2} \\ &\vdots \\ &= \pi_n\end{aligned}$$